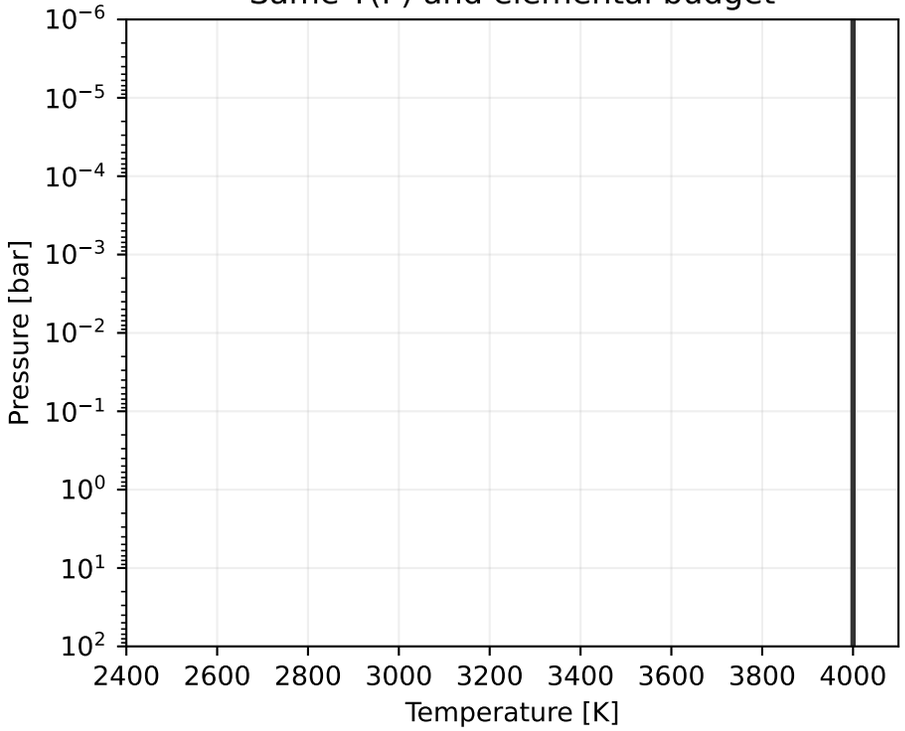
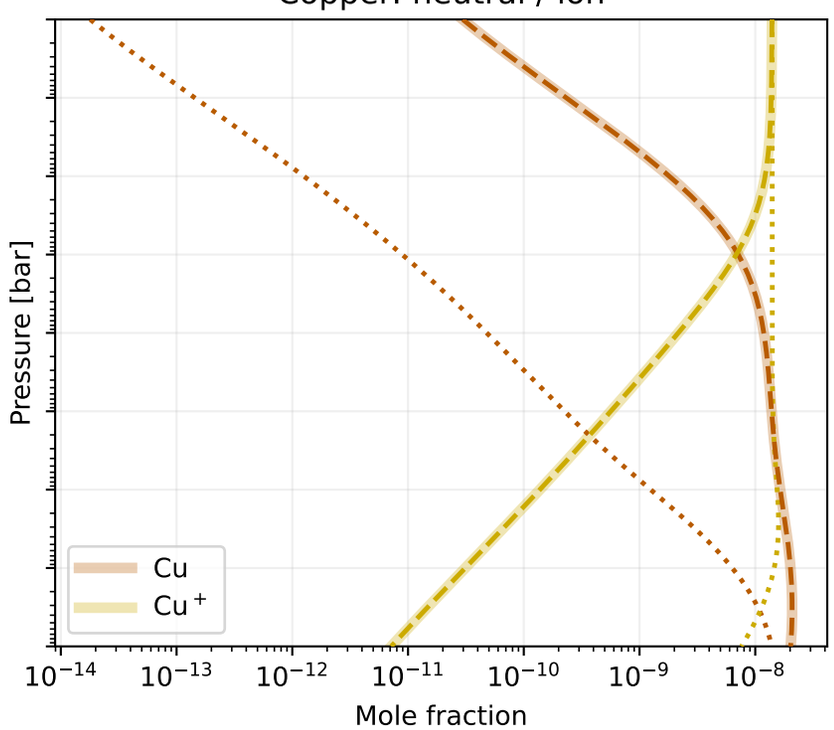
# 4000 K isothermal column | solar composition, same 537 species

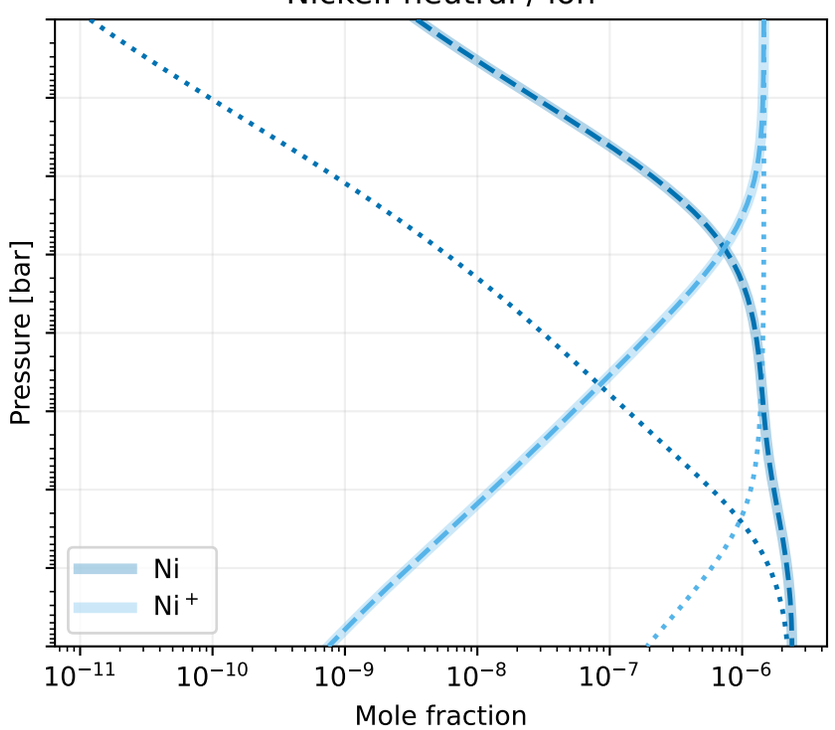
## Same T(P) and elemental budget



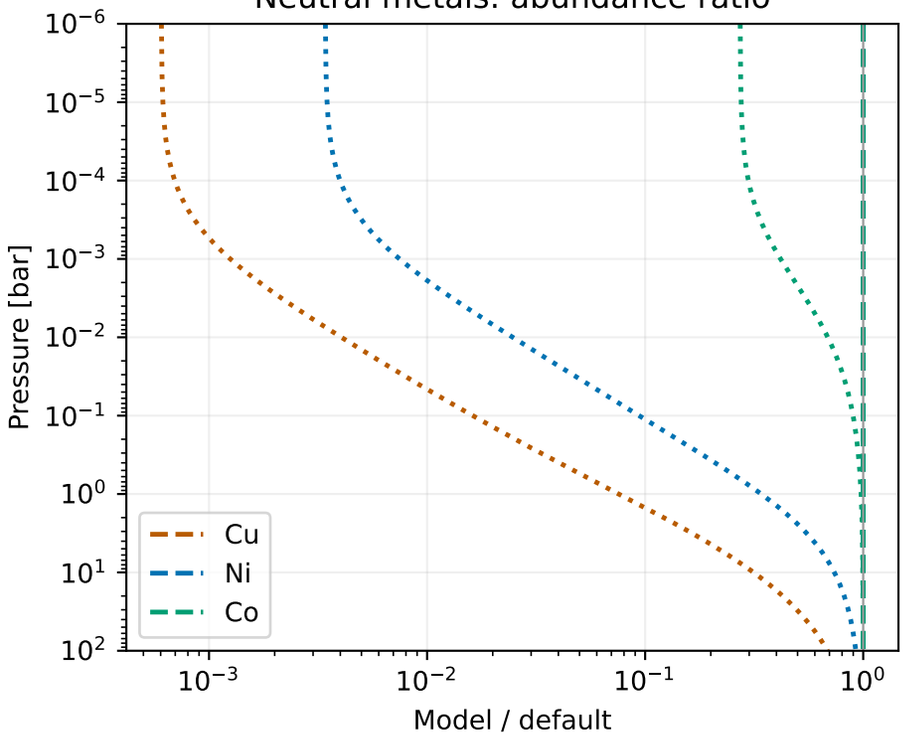

## Copper: neutral / ion



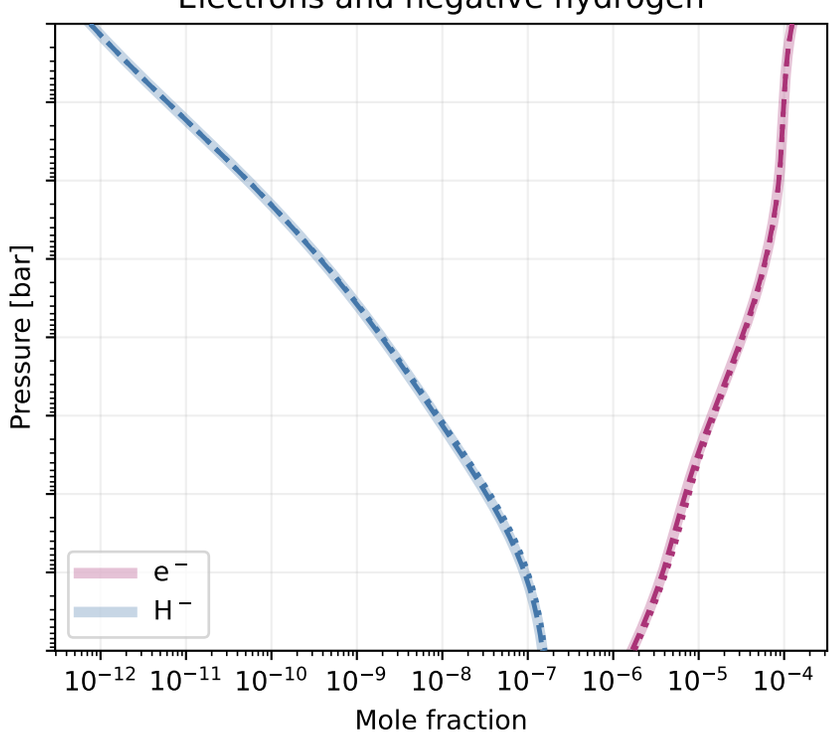

## Nickel: neutral / ion



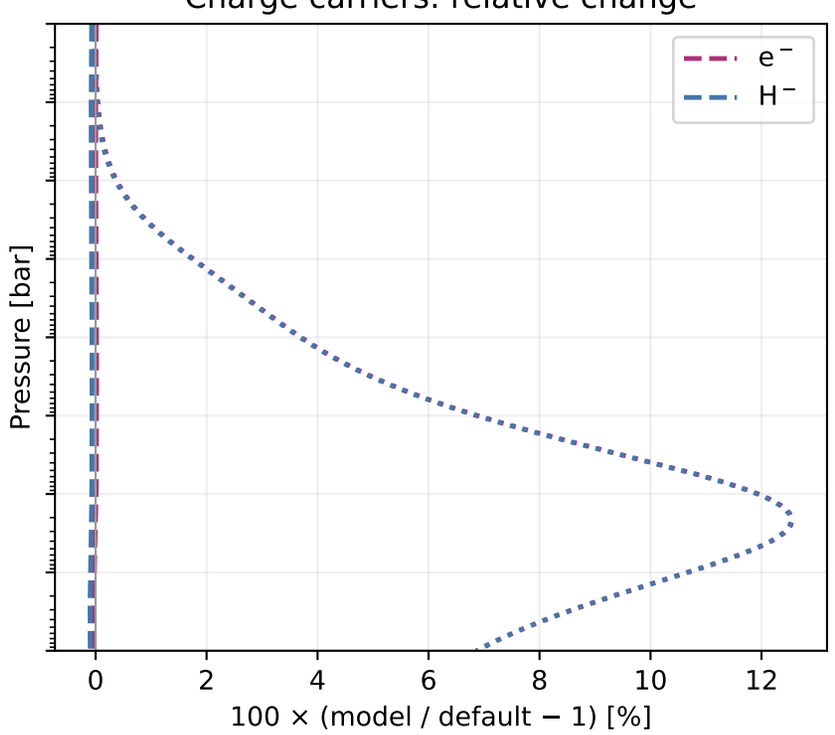

## Neutral metals: abundance ratio




## Electrons and negative hydrogen




## Charge carriers: relative change




ExoGibbs default (curated) — New FastChem data (preliminary) ····· Original FastChem 4 data

Ratio panels: dashed = new FastChem / default; dotted = original FastChem / default.
ExoGibbs solver for every curve; prescribed ideal-gas columns, without condensation or photoionization.